

CORONATION, AB, Canada

WMO: 713180

Lat: 52.0742N

Lon: 111.4494W

Elev: 2595

StdP: 13.37

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.9	(c) -21.1	(d) -35.1	(e) 0.8	(f) -26.2	(g) -28.5	(h) 1.2	(i) -20.7	(j) 29.4	(k) 19.0	(l) 26.4	(m) 17.4	(n) 7.2	(o) 270	(p) 0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	85.6	62.0	81.6	60.7	77.9	59.6	64.7	79.0	63.0	76.6	61.3	74.2	10.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	85.9	68.3	58.3	80.0	66.5	56.5	74.8	65.1	31.1	79.1	29.8	76.8	28.5	74.3	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.5	(b) 22.3	(c) 19.7	(e) -34.0	(f) 91.5	(g) 6.0	(h) 3.8	(i) -38.4	(j) 94.2	(k) -41.9	(l) 96.4	(m) -45.3	(n) 98.6	(o) -49.7	(p) 101.3		
(4) 25.5	(5) 22.3	(6) 19.7	(7) -34.0	(8) 68.1	(9) 5.8	(10) 2.2	(11) -38.2	(12) 69.6	(13) -41.5	(14) 70.9	(15) -44.8	(16) 72.1	(17) -49.0	(18) 73.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	8.4	11.1	21.9	37.7	49.4	56.9	62.5	60.7	51.3	38.1	22.5	10.9
	DBStd	22.35	15.94	13.88	14.26	9.73	7.82	5.87	5.72	6.29	8.09	9.04	12.58	14.62
	HDD50	6250	1290	1089	871	378	108	11	4	80	379	826	1213	
	HDD65	10633	1755	1509	1336	819	486	249	116	161	414	835	1275	1678
	CDD50	1167	0	0	0	9	88	218	389	336	118	9	0	0
	CDD65	76	0	0	0	0	1	6	39	28	2	0	0	0
	CDH74	1786	0	0	0	2	78	170	713	636	183	4	0	0
	CDH80	504	0	0	0	0	10	29	226	192	47	0	0	0
Wind	WSAvg	9.4	9.6	9.3	9.8	10.5	10.3	9.3	8.5	8.3	8.9	9.6	9.4	9.2
Precipitation	PrecAvg	15.9	0.8	0.7	0.8	0.9	1.6	2.5	2.9	2.0	1.6	0.6	0.6	0.9
	PrecMax	22.4	2.0	1.5	2.2	3.0	4.6	7.0	8.3	5.9	5.5	2.0	1.5	1.7
	PrecMin	9.4	0.2	0.1	0.1	0.0	0.2	0.6	0.6	0.2	0.1	0.1	0.1	0.2
	PrecStd	3.0	0.5	0.4	0.5	0.6	0.9	1.4	1.7	1.4	1.2	0.4	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	40.4	55.5	71.9	80.9	83.5	91.3	90.3	85.9	72.0	54.6	44.3
		MCWB	35.5	35.0	43.5	51.6	57.3	61.5	63.9	62.0	58.8	52.5	43.5	36.9
	2%	DB	37.7	36.6	48.6	65.8	75.8	78.7	86.1	85.1	79.1	63.8	47.2	39.6
		MCWB	33.7	32.5	39.3	48.4	54.4	59.5	63.8	61.4	57.1	49.1	39.5	34.4
	5%	DB	34.4	33.2	42.9	59.9	71.0	74.8	81.6	81.1	74.0	58.4	42.2	35.3
		MCWB	31.1	30.2	36.3	45.1	53.0	57.9	62.4	60.5	55.6	46.1	36.0	31.4
	10%	DB	30.3	30.0	38.3	54.8	66.6	71.2	77.4	76.8	68.5	53.1	37.9	31.1
		MCWB	27.9	27.5	33.6	42.4	50.4	56.4	61.2	59.6	53.2	42.9	33.5	28.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	35.8	44.8	52.2	59.4	64.3	68.2	65.8	60.7	53.9	44.1	37.8
		MCDB	40.0	39.6	53.8	69.6	77.7	77.8	83.0	79.7	78.3	69.1	53.0	43.3
	2%	WB	33.9	32.7	40.0	49.0	56.2	61.6	65.6	64.0	58.5	50.1	39.9	34.6
		MCDB	37.4	35.8	47.7	64.4	71.3	74.3	79.7	78.4	75.1	61.7	46.5	39.3
	5%	WB	31.2	30.3	36.6	46.0	54.0	59.6	63.8	62.3	56.3	46.9	36.5	31.5
		MCDB	34.2	33.1	42.1	58.7	68.4	70.9	77.8	76.5	71.9	56.6	41.1	35.0
	10%	WB	27.8	27.5	34.1	43.0	51.7	57.7	62.1	60.5	54.1	44.0	33.8	28.3
		MCDB	30.2	30.0	38.2	53.1	64.5	68.3	74.6	74.0	67.6	52.4	37.5	31.1
Mean Daily Temperature Range	5% DB	MDBR	17.7	18.4	17.7	22.0	24.4	22.4	24.8	25.5	25.5	22.4	17.8	17.6
		MCDBR	20.7	20.7	22.8	32.0	32.3	29.4	32.2	34.2	36.2	31.8	23.3	20.4
		MCWBR	17.3	17.4	15.0	17.1	15.6	13.3	13.1	14.1	16.9	18.1	16.0	15.9
	5% WB	MCDBR	20.7	20.3	21.5	30.6	29.3	25.2	28.2	29.0	32.9	30.0	21.5	19.7
		MCWBR	17.7	17.2	14.7	16.9	14.7	12.5	12.7	13.0	16.1	17.6	15.3	15.7
Clear-Sky Solar Irradiance	taub		0.266	0.309	0.288	0.336	0.359	0.368	0.359	0.374	0.319	0.283	0.258	0.252
	taud		2.300	2.208	2.449	2.343	2.324	2.341	2.406	2.334	2.459	2.540	2.471	2.325
	Ebn at Noon		237	253	292	286	283	280	281	270	276	268	245	223
	Edh at Noon		20	29	28	35	38	38	35	35	28	21	16	16
All-Sky Solar Radiation	RadAvg		359	695	1145	1521	1843	1905	2029	1650	1174	706	366	270
	RadStd		32	56	86	136	124	125	94	129	115	92	34	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page